BSc in Artificial Intelligence

A.A. 2025/-26

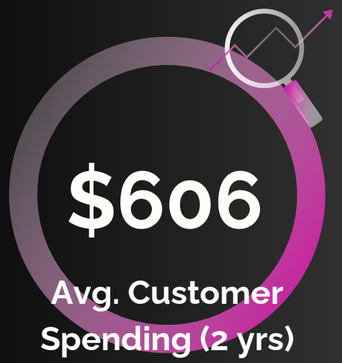
Exam Project 1:

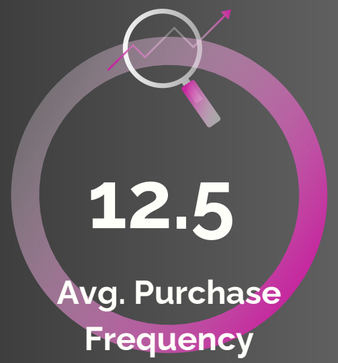
Customer Segmentation & Communication Strategy

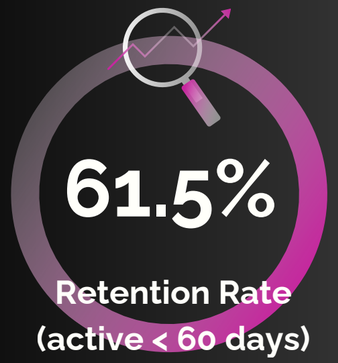
Made by
Lucrezia Bernini

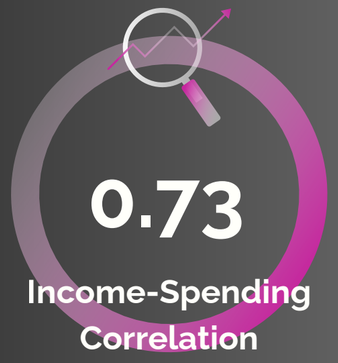
Customer Base Analysis

4 KPIs:









Hidden Trends Discovered:

1. "Smart Family Indulgence" — Families with teenagers are heavy deal-seekers (highest *NumDealsPurchases*), while families with small children cut luxury spending.
Campaign idea: bundled discounts on mid-tier wines and bulk products via targeted email.

2. Catalog Affinity — Customers who accept promotions buy primarily via Catalog (r ≈ 0.35).
Future promo campaigns should prioritise this channel.

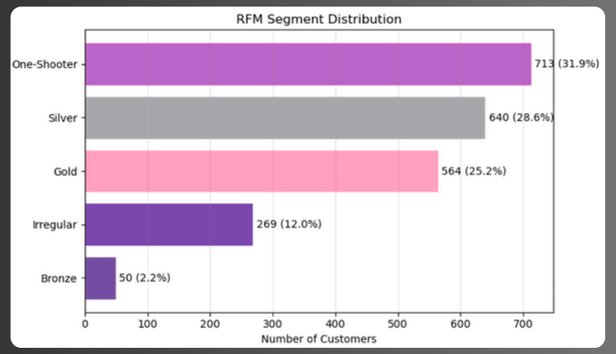
3. Window Shopper Effect — Web visits are negatively correlated with purchases (r = −0.42): massive traffic that doesn't convert.
Opportunity: scarcity tactics (Flash Sales) to drive quick conversion.

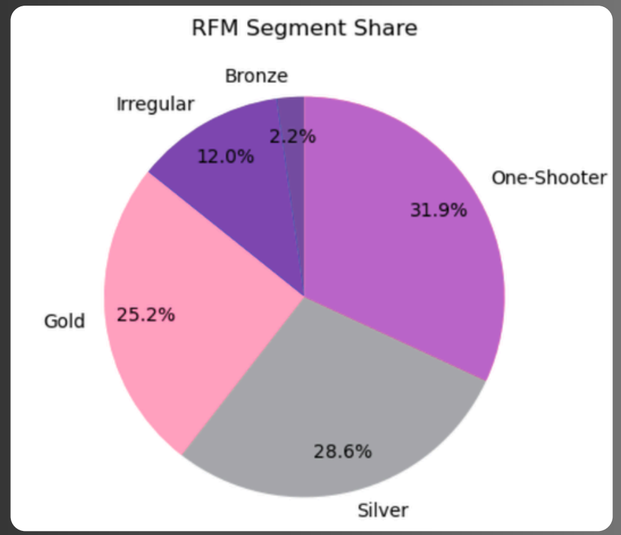
Purchase Channel Split: In-Store 46% | Web 33% | Catalog 22%

Dual-Lens Segmentation

RFM Heuristic Segments

Segment	Strategic Goal
Gold	Gratification
Silver	Spending Increase
Bronze	Spending Increase
One-Shooter	Frequency Increase
Irregular	Churn Prevention

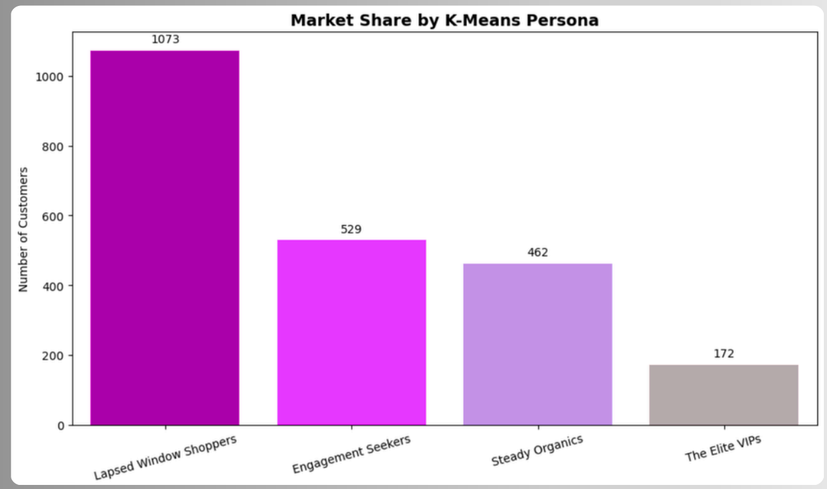




K-Means Behavioral Personas

K=4 (Elbow + Silhouette + Pseudo-F validation)

Persona	Customers
The Elite VIPs	172 (8%) — Highest spend & engagement
Engagement Seekers	529 (24%) — High promo acceptance, deal hunters
Lapsed Window Shoppers	1,073 (48%) — High web visits, low conversion
Steady Organics	462 (20%) — Normal spend, no promo reliance



Key Gap: K-Means captures promo sensitivity and web browsing that RFM misses — e.g. "Gold" customers might mask a subset of "Irregular" buyers who concentrate purchases in bursts.

Tailored Channel & Tone per Persona

Right message, right platform, right audience: maximise ROI through personalisation.

Segment	Channel	Message Tone	Goal
The Elite VIPs	Premium Catalogs + VIP Email	Exclusive, Appreciative	Retain & Reward
Engagement Seekers	Targeted Email + SMS Alerts	Urgent, Scarcity, FOMO	Convert Deals
Lapsed Window Shoppers	Exit-Intent Popups + Retargeting Ads	Re-engaging, Welcoming	Win Back & Convert
Steady Organics	Email Newsletters + Social Media	Educational, Trust-building	Nurture & Grow

Strategic Recommendation: Integrate both segmentation views into a **unified Customer.**

Strategy Matrix: use RFM to determine **priority** (*who to contact first*) and K-Means to determine **how** (*channel, message, offer type*). This dual-lens approach maximises both **efficiency** and **personalisation**.